



Homework No.2: Design of an IDS based on Rules Using SNORT (Teamwork).

OBJECTIVE

In this assignment, the student will get familiar with the design of an IDS based on rules using Snort [4, 5]. The objective is to identify several types of attacks, including Ping, DoS/DDoS, ARP spoofing, DNS Poisoning, traceroute, and Nmap scans.

PROCEDURE

1. TCP SYN Flood Attack

Firstly, TCP SYN flooding attack is a kind of **Denial-of-Service (DoS)** attack. This SYN flooding attack is using the weakness of TCP/IP. TCP is one of the core protocols of the Internet Protocol Suite. TCP is one of the two original components of the suite (the other being Internet Protocol, or IP), so the entire suite is commonly referred to as TCP/IP [1]. IP performs actual delivery of data via Internet, whereas TCP concerns only the data packets. TCP connection between the two end computers is established by a three-way handshake mechanism as follows [1].

Step 1. Client A sends a SYN packet to Client B.

Step 2. Client B sends a SYN/ACK packet to Client A.

Step 3. Client A sends a ACK packet to Client B.

TCP SYN Flood attack uses the three-way handshake mechanism. First client A, an attacker, sends a SYN packet to client B. Then client B sends a SYN/ACK packet to client A. As a normal three-way handshake mechanism client A should send an ACK packet to client B, however, client A does not send an ACK packet to client B. In this case client B is waiting for an ACK packet from client A. This status of client B is called “**half-open connection**”. This kind of incomplete connection is stored in Backlog Queue. After 75 seconds the incomplete connection is removed from Backlog Queue and it is disconnected [1]. However, if client A sends lots of SYN packets before client B removes incomplete connections from Backlog Queue, then Backlog Queue in client B is overflowed. In this case client B cannot accept TCP connection at all. This is called **DoS**, and this type of attack is TCP SYN Flood Attack.

To perform the TCP SYN Flood Attack you can use **Engage Packet Builder**, **Scapy**, **Low Orbit Ion Cannon (LOIC)**, **RUDY (R-U-Dead-Yet?)**, **hping3**, **DDoSIM** [2, 10, 11]. In this task, to detect this type of attack you can use either *Wireshark* [3] (define this type of rule in your IDS) or the **netstat** command. If you type the command, **netstat -n -p tcp**, then you can see if a large number of connections are in the SYN_RECEIVED state, so it is possible to conclude that the system is under attack. **Define the corresponding SNORT rules to identify these malicious network activities.**

2. ARP and DNS Poisoning

In this experiment, use the BetterCAP tool to perform ARP and DNS poisoning (that is, a substitution of false IP address at the DNS level) in LAN environment using VMware workstation in which you

have to install Kali Linux and *Ettercap tool* to sniff the local traffic in LAN. DNS Poisoning is quite similar to ARP. To initiate DNS poisoning, you have to start with ARP poisoning. Note that the *DNS spoof plugin* available in Ettercap should be used. Firstly, open up the terminal and type “nano etter.dns”. This file contains all entries for DNS addresses which are used by Ettercap to resolve the domain name addresses. In this file, please add a fake entry like “Facebook”. If someone wants to open Facebook, he/she will be redirected to another website. Now insert the entries under the words “Redirect it to www.linux.org”. The rest of the process is the same as Homework No. 1 [6-9]. **Define the corresponding SNORT rules to identify these network attacks.**

3. Network Mapping and PoD

The purpose of this second task is to gain understanding of some of the basic reconnaissance tools that are available to an outside attacker. Imagine that you are interesting in launching an attack against some organization. Assume you have already used Whois and found that the IP address range of the target organization is 10.24/16. Now you want to “case the joint” and gather as much information as you can about the target network. Use **traceroute**, **ping (tool: hping3)**, and **nmap** to determine as much as you possibly can about the secret network. This includes network layout, hosts, routers, switches and hubs, and how they are connected; IP addresses of hosts and router interfaces; open ports on the hosts; types and versions of routers; OS on each host, including OS version. **Define the corresponding SNORT rules to identify these network attacks.**

Note that each team must record a video in English, demonstrating participation from each student with the camera turned on. This video should explain the homework, and a **URL link to the video must be included in the report.**

REFERENCES

1. Jim Kurose, Keith Ross, *Computer Networking: A Top-Down Approach Featuring the Internet*, 4/e Addison Wesley, 2008.
2. Engage Security Builder URL <http://www.engagesecurity.com/products/engagepacketbuilder>
3. Wireshark URL <http://www.wireshark.org/>
4. Snort URL <https://www.snort.org/>
5. Suricata URL <https://suricata-ids.org/>
6. BetterCap URL <https://www.bettercap.org/>
7. EtterCap URL <https://www.ettercap-project.org/>
8. Kali URL <https://www.kali.org/>
9. URL <https://www.tutorialspoint.com/>
10. DDoSIM URL <https://ddosim.live/>
11. RUDY URL <https://www.invicti.com/learn/rudy-attack>

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